END-TO-END DATA ENGINEERING CAPSTONE

Session 1

Project Scope, Problem Statement & Architecture Design

Facilitator Guide · 60 Minutes

Platform: Databricks Free Edition (Lakehouse + Unity Catalog)
Dataset: Olist Brazilian E-Commerce (Kaggle)
Architecture: Medallion · Bronze ⟶ Silver ⟶ Gold

1. Session at a Glance

This is the foundation session of a five-part capstone. Before anyone writes a line of ingestion code, learners must be able to answer three questions: **what problem are we solving, what does the data look like, and how will the platform be shaped to deliver the answer.** S1 establishes that framing so every later session — modelling, ETL, dashboards, presentation — has a clear north star.

Learning outcomes — by the end of this hour, participants can:

- Translate a vague business ask into a written, measurable problem statement.
- Define project scope with explicit in-scope and out-of-scope boundaries.
- Explain the medallion (Bronze → Silver → Gold) architecture and justify why each layer exists.
- Draw the end-to-end data flow from source system to BI dashboard.
- Identify where AI capabilities fit into the pipeline and why.

Where the whole capstone is going

Session	Focus	Key Deliverable
S1	Scope, problem, architecture	Project charter + architecture diagram
S2	ER design + SQL Server DDL	Physical data model with constraints
S3	Databricks Bronze→Silver→Gold ETL	Working medallion pipeline job
S4	Power BI dashboards	Executive + operational dashboards
S5	Presentation + Q&A + CBT	Final defense & assessment

2. One-Hour Timing Plan

Use this as the running order. The design keeps the group in discussion for the first half and in architecture for the second, ending with a scope-sign-off that becomes the input to S2.

Time	Segment	Facilitator Activity
0–5 min	Framing	Set the scene: the marketplace, the messy CSVs, the ask from leadership.
5–15 min	Problem statement	Co-write the problem statement live; contrast weak vs strong versions.
15–25 min	Scope & objectives	Define in-scope / out-of-scope; convert goals to measurable KPIs.
25–30 min	Source-to-target overview	Walk the source system and the target lakehouse at a high level.
30–45 min	Medallion architecture	Explain Bronze / Silver / Gold; justify each layer's purpose.
45–53 min	AI & BI touchpoints	Show where AI functions and Power BI plug in.
53–60 min	Scope sign-off + preview	Confirm the charter; preview S2 (ER modelling).

3. The Business Context

The scenario is deliberately realistic. **Olist** is a Brazilian marketplace that connects thousands of small sellers to customers across the country. Orders, payments, shipping, and reviews are all captured — but in nine disconnected operational CSV files, each telling only part of the story.

Leadership's ask (the raw, unstructured version):

> **Verbatim stakeholder request**
> *"We're growing fast but flying blind. I want to know how much we're selling, whether deliveries are actually on time, why customers leave bad reviews, and which sellers and categories drive the business. Can you get me one place to see all of this?"*

Notice what's wrong with this as an engineering brief: it's a wish, not a specification. **The facilitator's job in the next segment is to turn it into something buildable.**

4. Writing the Problem Statement

A problem statement is a single, testable paragraph. Teach the contrast directly — show a weak one, then sharpen it live with the group.

Weak version (what to avoid)

> **Too vague to build from**
> *"Build a dashboard to help Olist understand its business better."*
> Problems: no owner, no metric, no boundary, no definition of success.

Strong version (the target)

> **Specific, measurable, bounded**
> Olist lacks a unified analytics platform, leaving revenue, delivery performance, and customer-satisfaction data trapped across nine operational files. This project will consolidate those sources into a governed lakehouse and deliver dashboards that report monthly revenue, on-time delivery rate, review-score drivers, and top categories and sellers — enabling data-driven operational decisions.

The strong version passes four tests — walk through each with the group:

- **Specific** — names the exact sources and outputs.
- **Measurable** — lists concrete metrics (revenue, on-time %, review drivers).
- **Bounded** — consolidation + dashboards, not ML forecasting or real-time streaming.
- **Outcome-oriented** — ends in a decision-enabling capability, not just a chart.

5. Scope & Measurable Objectives

Scope protects the project from expanding endlessly. Make the boundaries explicit and visible.

In scope vs out of scope

In Scope	Out of Scope
Batch ingestion of 9 Olist CSV sources	Real-time / streaming ingestion
Medallion cleansing & modelling (star schema)	Predictive ML models / forecasting
AI enrichment of review text (sentiment, category)	Custom-trained NLP models
Revenue, delivery, review, seller/category KPIs	Financial forecasting or budgeting
Power BI executive & operational dashboards	Mobile app or embedded portal

Turning goals into KPIs

Every objective must become something the pipeline can compute. This table becomes the acceptance criteria for S3 and S4.

Business Goal	Measurable KPI	Source Tables
Track sales performance	Monthly revenue, order count	orders, order_items
Assess delivery reliability	On-time delivery %, avg delivery days	orders
Understand dissatisfaction	Review score dist., AI issue category	order_reviews
Find top performers	Revenue by category / seller	products, sellers, order_items
Map the customer base	Orders by state / city	customers, geolocation

6. Architecture Design

With the problem and scope fixed, the architecture answers *how*. The design is a governed lakehouse built on the medallion pattern, moving data through three quality tiers before it reaches the dashboard.

End-to-end data flow

```
Source → Lakehouse → BI
Kaggle CSVs → Volume (raw landing)
     | batch ingest
BRONZE → SILVER → GOLD (Delta tables, Unity Catalog)
 raw as-is   cleaned/conformed   business-ready star schema
     |
Power BI (executive + operational dashboards)
```

Note: a rendered version of this diagram is included in the slide deck. Draw it live on the whiteboard as you narrate the flow — the act of drawing it is what makes it stick.

Why medallion? Justify each layer

The single most important teaching point of S1: each layer exists for a specific engineering reason. Never let learners see it as three arbitrary copies of the data.

Layer	Purpose	What Lives Here
Bronze	Auditability & replay	Raw ingest, exactly as-is, plus load metadata (ingest time, source file)
Silver	Quality & conformance	Typed, deduplicated, cleansed, business-conformed tables; AI-enriched reviews
Gold	Performance & semantics	Star schema (fact + dims) and pre-aggregated KPIs ready for BI

The one-line rationale to repeat: *Bronze is your safety net, Silver is your source of truth, Gold is your storefront.*

Platform components

- **Databricks Free Edition** — serverless compute, no cluster management.
- **Unity Catalog** — governance layer; catalog → schema → table/volume hierarchy.
- **Volumes** — governed storage for the raw Kaggle files before ingestion.
- **Delta Lake** — the table format underneath every layer; ACID, time-travel, schema enforcement.
- **Databricks Jobs** — orchestrates Bronze → Silver → Gold as dependent tasks.

Where AI fits

A differentiator of this capstone: native AI functions run directly in SQL on the review text, with no separate model to train or host.

- **ai_analyze_sentiment** — classifies each review comment as positive / negative / mixed.
- **ai_classify** — tags each review into an issue category (delivery, product quality, price, service).

The payoff, surfaced in S4: compare AI-derived sentiment against the numeric review score to find **what actually drives dissatisfaction** — the kind of insight a raw star rating alone can't give.

7. Scope Sign-Off & Handover to S2

Close the session by confirming the charter as a group. This checklist is the exit ticket — everything must be agreed before modelling begins.

1. Problem statement is written, specific, and measurable.
2. In-scope and out-of-scope lists are explicit and accepted.
3. Every business goal maps to at least one computable KPI.
4. The medallion architecture and data flow are understood by all.
5. AI and BI touchpoints are identified in the design.

Next session (S2): we translate this architecture into a physical data model — an ER diagram and SQL Server DDL with primary keys, foreign keys, constraints, and indexes. The KPI-to-source table map from Section 5 is the direct input to that design.

Facilitator notes

Running this session well

• Keep segment 4 interactive — co-write the problem statement on screen rather than presenting a finished one.

• The medallion justification (Section 6) is the concept most likely to be examined in S5 — spend real time on the ‘why’, not just the ‘what’.

• Have the architecture diagram ready to draw live; the whiteboard version outperforms any slide.

• End on time with a signed-off charter — an unclosed scope discussion derails S2.